

1 The off-line counterfactual inside the original holonomy construction

This calculation uses the original theta source in the final analytic continuation of the programme. In particular, the scalar measure below is computed from g/h with $g = 2\xi$; it is not a Gamma reference measure. The source equations being continued are (H5)–(H17), (H24)–(H28), and (H53)–(H55) of *All-holonomy descent of the original theta quotient*. The complete source record and locators are recorded in the accompanying audit. The conclusions concern the stated counterfactual and do not assert the existence of an off-line zero.

1.1 The retained arithmetic objects and every phase

Suppose $1/2 + \delta + i\gamma$ is a zero of g of its full order m , where $0 < \delta < 1/2$ and $\gamma > 0$. Reflection and conjugation give the complete quartet, with the same order. Retain

$$h(s) = \prod_{\epsilon, \eta \in \{-1, 1\}} (s - 1/2 - \epsilon\delta - i\eta\gamma)^m, \quad v_h = g/h, \quad w_h(u) = \frac{|v_h(1/2 + iu)|^2}{2\pi}. \quad (\text{HT1})$$

The division is entire and its full jet $v_h = j_h(v_h)$ is a unit. For an integer $k \geq 2$, put

$$c = k/2, \quad \ell_k = 1 + k(m-1), \quad q = \ell_k(k+1)^2, \quad m_{h,k} = w_h^{*k},$$

$$\chi(S) = \prod_{a,b=0}^k (S - c - (2a-k)\delta - i(2b-k)\gamma)^{\ell_k}, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S. \quad (\text{HT2})$$

These are the original cyclic sum quotient, its full multiplicities, and its multiplication operator. For completeness, in each tensor local factor the nilpotent sum is $n_1 + \dots + n_k$ with $n_i^m = 0$. Its power $k(m-1)$ has nonzero coefficient $(k(m-1))!/((m-1)!)^k$ on $n_1^{m-1} \dots n_k^{m-1}$, whereas the next power vanishes. Thus its nilpotency index is exactly ℓ_k . Every pair (a, b) is realized by choosing independently the real and imaginary signs in the k tensor factors. Distinct pairs give distinct complex sums, so the annihilator in the sum variable is exactly χ .

The original analytic construction provides F_h with Mellin transform v_h , and the original source and jet maps are

$$\mathcal{V}P = P(D_1 + \dots + D_k)F_h^{\otimes k}, \quad J^{(k)}\mathcal{V} = \eta\pi_\chi, \quad \eta[P] = v_h^{\otimes k}P(A_k)1. \quad (\text{HT3})$$

Here η is an injective complex linear module map. Its unit factor is retained. The norm of this source is exactly $\int_{\mathbb{R}} |P(c + iu)|^2 m_{h,k}(u) du$. The half-density map is

$$(\mathcal{U}_k F)(r, z) = e^{(kr + \sum_{i < k} z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r), \quad r = \log x_k, \quad z_i = \log x_i - \log x_k. \quad (\text{HT4})$$

It maps into $L^2(\mathbb{R}, dr; \mathcal{K})$ with $\mathcal{K} = L^2(\mathbb{R}^{k-1}, dz)$ and intertwines $D_1 + \dots + D_k$ with $-\partial_r + c$. No relative variable is removed.

Fix any $L > 0$ and $\theta \in [0, 2\pi)$. The source of phase θ is

$$(\mathcal{Z}_{L,\theta}\psi)(r) = \sum_{j \in \mathbb{Z}} e^{ij\theta} \psi(r + jL), \quad \psi(r + L) = e^{-i\theta} \psi(r) \quad \text{in the target domain.} \quad (\text{HT5})$$

On the original finite source, its Gram form is exactly the polynomial norm for the measure

$$\nu_{L,\theta} = \frac{2\pi}{L} \sum_{n \in \mathbb{Z}} m_{h,k}(u_n) \delta_{u_n}, \quad u_n = \frac{2\pi n + \theta}{L}, \quad S = c + iu. \quad (\text{HT6})$$

The factor $2\pi/L$ is unchanged. The measure has positive mass at every u_n . Indeed the nonzero entire function w_h has only isolated zeros on the line; hence $w_h > 0$ almost everywhere. For every u , the product $w_h(v)w_h(u-v)$ is positive almost everywhere in v , so its integral is positive. Iterated convolution preserves strict positivity. The original weighted Schwartz estimates give convergence and all moments in (HT6). A nonzero polynomial cannot vanish at every lattice point; consequently every finite polynomial Gram matrix is positive definite.

For $N \geq q-1$, let $P_N = \mathbb{C}[S]_{\leq N}$ and $J_N = \pi_\chi|_{P_N}$. With the fixed coefficient bases define

$$M_{N,\theta} = (\mathcal{Z}_{L,\theta}\mathcal{U}_k\mathcal{V}|_{P_N})^*(\mathcal{Z}_{L,\theta}\mathcal{U}_k\mathcal{V}|_{P_N}), \quad K = J_N M_{N,\theta}^{-1} J_N^*, \quad G = K^{-1},$$

$$C_{N,\theta} = M_{N,\theta}^{-1} J_N^* G, \quad R = \mathcal{Z}_{L,\theta}\mathcal{U}_k\mathcal{V} C_{N,\theta}. \quad (\text{HT7})$$

Thus $J_N C_{N,\theta} = I_E$ and $R^* R = G$. All inverses in (HT7) exist: $M_{N,\theta}$ is positive definite and J_N is surjective. No inverse at degree $q-2$ is used.

1.2 A uniform phasewise lower bound for the original control

Define

$$H = G^{-1/2}(A^*G + GA - kG)G^{-1/2}, \quad \mathcal{L}_{h,k} = 2\delta\ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \quad (\text{HT8})$$

For every $k \geq 2$, $L > 0$, $\theta \in [0, 2\pi)$, and $N \geq q-1$, the nonzero eigenvalues of H are $\varepsilon, -\varepsilon$, and

$$\boxed{\varepsilon \geq \mathcal{L}_{h,k} > 0, \quad \frac{\mathcal{L}_{h,k}}{q} \geq \frac{\delta k}{2}.} \quad (\text{HT9})$$

Proof. Let $p_j(S)$ be the monic orthogonal polynomials for (HT6), with squared norms $\omega_j > 0$, and let $b_j = [p_j] \in E$. In the real coordinate u , write $p_j(c + iu) = i^j r_j(u)$, where r_j is real monic. Orthogonality and the symmetry of multiplication by u show

$$ur_j = r_{j+1} + a_j r_j + \frac{\omega_j}{\omega_{j-1}} r_{j-1}, \quad a_j \in \mathbb{R},$$

with the last term omitted at $j=0$. To verify that no earlier term occurs, pair ur_j with r_i , $i \leq j-2$, and move u to r_i ; its degree is at most $j-1$. Pairing with r_{j-1} gives precisely ω_j/ω_{j-1} . In the S coordinate this is

$$Sp_j = p_{j+1} + (c + ia_j)p_j - \frac{\omega_j}{\omega_{j-1}} p_{j-1}. \quad (\text{HT10})$$

Since the source Gram is diagonal in this basis, $K = \sum_{j=0}^N b_j b_j^* / \omega_j$. Apply A on the left and A^* on the right and use (HT10). The imaginary diagonal terms cancel, and each adjacent term except the last cancels its predecessor. This gives, including $N = q-1$,

$$AK + KA^* - kK = \frac{b_{N+1}b_N^* + b_N b_{N+1}^*}{\omega_N}. \quad (\text{HT11})$$

Congruence by $G^{1/2}$ identifies the left side with H . Thus its rank is at most two. Also $\text{Tr } H = 2 \text{Re } \text{Tr } A - kq = 0$ by the complete grid (HT2).

Let E_+ be the direct sum of the complete primary spaces with $a > k/2$, and let P be the orthogonal projection onto $G^{1/2}E_+$ in the standard Hermitian space. The latter subspace is invariant under $G^{1/2}AG^{-1/2}$. Taking traces in an orthonormal basis of this subspace gives

$$\text{Tr}(PH) = 2 \sum_{a > k/2} \sum_{b=0}^k \ell_k(2a-k)\delta = 2\delta\ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor = \mathcal{L}_{h,k}. \quad (\text{HT12})$$

The finite sum is r^2 if $k = 2r - 1$, and $r(r + 1)$ if $k = 2r$. A Hermitian rank-at-most-two matrix of trace zero has eigenvalues $\varepsilon, -\varepsilon$ and zeroes. For every orthogonal projection, $\text{Tr}(PH) \leq \varepsilon$: in an eigenbasis the positive contribution is at most ε and the negative one is nonpositive. Equation (HT12) proves the first inequality of (HT9), and $4\lfloor(k + 1)^2/4\rfloor \geq k(k + 1)$ proves the second. $\square$

1.3 The exact energy forced into the retained theta boundary

In $\mathcal{H}_{L,\theta} = L^2([0, L], dr; \mathcal{K})$ put $D_{L,\theta} = -\partial_r + c$, with domain $\{f \in H^1([0, L]; \mathcal{K}) : f(L) = e^{-i\theta} f(0)\}$. The domain of its square is $\{f \in H^2([0, L]; \mathcal{K}) : f(L) = e^{-i\theta} f(0), \quad f'(L) = e^{-i\theta} f'(0)\}$. The original smooth periodized source columns belong to these domains. Monic division in the original coefficient source gives the unique polynomial map $Q_{N,\theta}$ with

$$SC_{N,\theta} - C_{N,\theta}A = \chi Q_{N,\theta}.$$

Indeed applying J_{N+1} gives zero. Define the actual boundary

$$\mathcal{B} = \mathcal{Z}_{L,\theta} \mathcal{U}_k \mathcal{V}(\chi Q_{N,\theta}) = D_{L,\theta}R - RA. \quad (\text{HT13})$$

This retains its theta primitive. Explicitly, tensor monic division gives

$$\chi(S)Q(S) = \sum_i h(s_i)Q_i(\mathbf{s}),$$

because its image in the full tensor jet algebra vanishes. In the original theta complex, ϕ_h has degree zero, F_h has degree one, and $h(D)F_h = \Theta\phi_h$. The degree- $(k - 1)$ tensor primitive is exactly

$$\sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) (F_h^{\otimes(i-1)} \otimes \phi_h \otimes F_h^{\otimes(k-i)}).$$

The tensor differential contributes a further sign $(-1)^{i-1}$ when it acts on ϕ_h , because the preceding $i - 1$ factors have degree one. The two signs cancel, so its differential is precisely $\mathcal{V}(\chi Q)$. Applying the same half-density and phase maps gives precisely (HT13).

Set

$$U = RG^{-1/2}, \quad T = \mathcal{B}G^{-1/2}, \quad Z = U^*T, \quad T_\perp = (I - UU^*)T.$$

Then the following exact energy identity, with no omitted source term, holds:

$$\boxed{\text{Tr}(G^{-1}\mathcal{B}^*\mathcal{B}) = \frac{\varepsilon^2}{2} + \left\| \frac{Z - Z^*}{2} \right\|_{\text{HS}}^2 + \|T_\perp\|_{\text{HS}}^2 \geq \frac{\mathcal{L}_{h,k}^2}{2} \geq \frac{\delta^2 k^2 q^2}{8}.} \quad (\text{HT14})$$

All quantities on the left are calculated from g/h and its original theta representatives. In particular the same estimate holds after integration with the original Fourier inversion measure:

$$\int_0^{2\pi} \text{Tr}(G_{N,\theta}^{-1} \mathcal{B}_{N,\theta}^* \mathcal{B}_{N,\theta}) \frac{d\theta}{2\pi} \geq \frac{\mathcal{L}_{h,k}^2}{2}. \quad (\text{HT15})$$

Proof. The two arguments in integration by parts have the same boundary multiplier $e^{-i\theta}$, so the endpoint products cancel. Hence $D_{L,\theta}^* + D_{L,\theta} = 2cI$ on the stated domain. Inserting $D_{L,\theta}R = RA + \mathcal{B}$ and $R^*R = G$ gives

$$A^*G + GA - kG = -(R^*\mathcal{B} + \mathcal{B}^*R), \quad H = -Z - Z^*.$$

Since $U^*U = I$, the decomposition $T = UZ + T_\perp$ is orthogonal in Hilbert–Schmidt norm. The Hermitian and skew-Hermitian parts of Z are orthogonal for the real Hilbert–Schmidt scalar product. Therefore

$$\|T\|_{\text{HS}}^2 = \frac{1}{4}\|H\|_{\text{HS}}^2 + \left\|\frac{Z - Z^*}{2}\right\|_{\text{HS}}^2 + \|T_\perp\|_{\text{HS}}^2.$$

By (HT9), $\|H\|_{\text{HS}}^2 = 2\varepsilon^2$, which proves (HT14). For fixed k, L, N the weighted source sums and their derivatives converge uniformly in θ . Thus $M_{N,\theta}$ is continuous. Its smallest eigenvalue is positive on the compact phase circle, so its inverse, $G_{N,\theta}$, and the boundary columns are continuous there. Integration of (HT14) proves (HT15). $\square$

1.4 The full lattice test for the original Laplacian boundary

Put $\mathcal{L}_{L,\theta} = D_{L,\theta}^2 - kD_{L,\theta} = \partial_r^2 - c^2$. Direct substitution of (HT13) gives the exact map

$$\mathcal{L}_{L,\theta}R - R(A^2 - kA) = (D_{L,\theta} - k)\mathcal{B} + \mathcal{B}A. \quad (\text{HT16})$$

If $Av = \lambda v$ with $v \neq 0$, write $\lambda = c + x + iy$. Then

$$\frac{\|((D_{L,\theta} - k)\mathcal{B} + \mathcal{B}A)v\|}{\|Rv\|} \geq \min_{n \in \mathbb{Z}} \sqrt{(u_n^2 + x^2 - y^2)^2 + 4x^2y^2}. \quad (\text{HT17})$$

In particular every phase and every admitted degree satisfy, on an eigenvector for $\lambda = k(1/2 + \delta + i\gamma)$,

$$\boxed{\frac{\|((D_{L,\theta} - k)\mathcal{B} + \mathcal{B}A)v\|}{\|Rv\|} \geq 2k^2\delta\gamma.} \quad (\text{HT18})$$

For an eigenvalue with $x \neq 0$ and $y = 0$, (HT17) also gives the positive lower bound $x^2 + \min_n u_n^2$, even though its imaginary part is zero.

Proof. The Fourier basis $L^{-1/2}e^{-iu_nr}$ diagonalizes the quasi-periodic Laplacian with eigenvalues $-u_n^2 - c^2$, each acting identically on $\mathcal{K}$. The source vectors lie in its domain by the original weighted differentiability estimates. Writing Rv in that basis and using Parseval gives

$$\|(\mathcal{L}_{L,\theta} - (\lambda^2 - k\lambda))Rv\|^2 = \sum_n ((u_n^2 + x^2 - y^2)^2 + 4x^2y^2) \|\widehat{Rv}(n)\|_{\mathcal{K}}^2.$$

The lattice is discrete and the expression tends to infinity as $|n| \rightarrow \infty$, so its infimum is a minimum. Also R is injective because $R^*R = G \succ 0$. Divide the displayed identity by $\|Rv\|^2$; (HT16) proves (HT17). The extreme primary factor in (HT2) is nonzero and has an eigenvector, so substituting $x = k\delta, y = k\gamma$ proves (HT18). If $y = 0$, the square root in (HT17) is $u_n^2 + x^2$. $\square$

1.5 What these computations decide

The original off-line counterfactual survives passage to every finite holonomy quotient only with the explicit boundary energy (HT14) and Laplacian residual (HT17). These quantities belong to the original arithmetic source. The statement is compatible with the exact all-phase identity $\int M_{N,\theta} d\theta / (2\pi) = M_N$ and with the original small gluing-norm estimate. That estimate concerns $C_N - C_{N,\theta}$; (HT13) applies S and subtracts right multiplication by A , preserving a different, explicitly displayed source quantity. There is no claim here that averaging the $G_{N,\theta}$ preserves the rank-two identity (HT11): the lower bound (HT15) integrates the proved phasewise energies themselves.

The transcript's threshold-four volume lower bound was already present before its final Deligne comparison. Its last counterexamples retain many generic identities but do not satisfy the actual divisor condition $h \mid 2\xi$. Consequently they disprove the reduced generic inference stated there; they do not disprove the entire arithmetic programme or provide an RH counterexample. Equations (HT9), (HT14), and (HT17) give further quantified consequences within the original programme without replacing its theta metric or its represented relations.